Engineering Document E14 — Memory Engine & Knowledge Architecture

Product Name: HQ

Engineering Package: Version 1.0

Purpose

This document defines the Memory Engine that gives HQ long-term intelligence.

Unlike traditional AI assistants that forget context after a conversation, HQ maintains structured organizational memory that enables executives to reason consistently, learn from previous work, and improve over time.

Memory is the foundation of organizational intelligence.

Vision

HQ should remember like a real executive team.

Executives should remember:

- Previous decisions
- Organization goals
- Brand identity
- Mission outcomes
- Lessons learned
- User preferences
- Collaboration history

Memory transforms isolated AI responses into continuous organizational knowledge.

Memory Principles

The Memory Engine must be:

- Persistent
- Hierarchical
- Searchable
- Secure
- Context-aware
- Versioned
- Permission-aware
- Efficient

Every memory must have a clear purpose.

Memory Architecture

```
User Memory
   |
Organization Memory
   |
Executive Memory
   |
Mission Memory
   |
Working Memory
   |
Knowledge Library
   |
Vector Search Layer
   |
Prompt Composition Engine
```

The CEO retrieves only the memory required for the current mission.

Memory Layers

HQ organizes memory into specialized layers.

Layer 1 — User Memory

Stores:

- Preferred language
- Writing style
- Communication tone
- Favorite workflows
- Frequently used templates
- Personal preferences

Example:

"The user prefers concise executive summaries."

Layer 2 — Organization Memory

Stores:

- Company profile
- Mission statement
- Brand voice
- Products
- Services
- Goals
- Policies
- Organizational structure

Every executive can reference organization memory.

Layer 3 — Executive Memory

Each executive owns private domain knowledge.

Marketing remembers:

- Campaign performance
- Brand messaging
- Audience insights

Finance remembers:

- Budget history
- Revenue trends
- Cost optimization

Engineering remembers:

- Architecture decisions
- Technical debt
- Deployment history

Executives do not automatically access each other's private memories.

Layer 4 — Mission Memory

Every mission stores:

- Mission brief
- Executive assignments
- Decisions
- Deliverables
- Revisions
- Final outputs
- Lessons learned

Mission memory remains available for future reference.

Layer 5 — Working Memory

Temporary memory used during execution.

Contains:

- Active tasks
- Intermediate reasoning
- Current collaboration
- Temporary calculations

Working memory expires after mission completion unless promoted.

Layer 6 — Knowledge Library

Stores uploaded knowledge.

Examples:

- SOPs
- PDFs
- Brand Guidelines
- Product Documentation
- Policies
- Research Reports
- Meeting Notes
- Technical Manuals

Documents become searchable organizational knowledge.

Memory Lifecycle

```
Capture

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Validate

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Categorize

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Store

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Index

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Retrieve

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Update
```

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```
Archive
```

Memory follows a complete lifecycle.

Memory Capture

Sources include:

- User input
- Executive outputs
- Mission results
- Uploaded files
- Organization settings
- Analytics
- Feedback

Only valuable information is stored.

Memory Classification

Every memory receives metadata.

Attributes:

- Type
- Owner
- Organization
- Executive
- Mission
- Priority
- Tags
- Timestamp
- Expiration
- Security Level

Classification improves retrieval accuracy.

Memory Retrieval

Retrieval follows this priority:

Working Memory

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Mission Memory

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Executive Memory

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Organization Memory

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Knowledge Library

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User Memory

Only relevant memories are injected into prompts.

Vector Search

Knowledge retrieval uses semantic search.

Capabilities:

- Natural language search
- Similarity search
- Context expansion

- Related document discovery

This enables Retrieval-Augmented Generation (RAG).

Memory Promotion

Not every memory becomes permanent.

Rules:

Temporary

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Useful

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Frequently Referenced

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Long-Term Memory

Executives promote valuable knowledge automatically, subject to review.

Memory Expiration

Examples:

Working Memory:

Expires after mission completion.

Temporary notes:

Expire after a configurable period.

Organization Memory:

Persistent until updated.

Retention policies are configurable.

Memory Permissions

Access follows organizational security.

Users access:

- Their own preferences
- Organization data they are authorized to view

Executives access:

- Their domain memory
- Shared organizational knowledge
- Assigned mission memory

Sensitive memories require explicit authorization.

Knowledge Library

Documents are processed through:

Upload

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Validation

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Text Extraction

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Chunking

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Embedding

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Indexing

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Search

↓

Retrieval

Knowledge becomes immediately available for future missions.

Memory Versioning

Every update records:

- Previous value
- New value
- Editor
- Timestamp
- Reason

Historical memory is never silently overwritten.

Learning Loop

After each completed mission:

The CEO evaluates:

- Success
- Mistakes
- Revisions

- User feedback
- Executive performance

Approved lessons become organizational knowledge.

Analytics

Track:

- Retrieval accuracy
- Memory usage
- Search latency
- Knowledge growth
- Frequently referenced memories
- Executive learning trends

These metrics guide optimization.

Security

Memory security includes:

- Encryption at rest
- Encryption in transit
- Role-based access
- Audit logging
- Secure deletion
- Organization isolation

No organization can access another organization's memory.

AI Integration

The Memory Engine integrates with:

- CEO Executive
- Executive Library
- Prompt Composition Engine
- Mission Orchestrator
- Analytics Center

Memory powers every intelligent decision.

Future Evolution

Supports:

- Cross-document reasoning
- Automatic knowledge summarization
- Voice transcript memory
- Image understanding
- Video indexing
- Knowledge graphs
- Enterprise knowledge connectors

The architecture is designed for continuous expansion.

Success Criteria

The Memory Engine succeeds when:

- Executives remember relevant organizational knowledge.
- Memory retrieval is fast and accurate.
- Token usage is minimized through selective context.
- Learning improves mission quality over time.
- Security and permissions are consistently enforced.

Conclusion

The HQ Memory Engine transforms isolated AI interactions into a persistent organizational intelligence system.

By organizing knowledge into hierarchical memory layers, integrating semantic retrieval, enforcing permissions, and continuously learning from mission outcomes, HQ creates an AI Executive Headquarters that grows more knowledgeable and effective with every mission. This architecture enables long-term continuity, stronger collaboration, and a truly intelligent executive organization.